EXP 13 : IMPLEMENTATION AND PERFORMANCE EVALUATION OF CNN-BASED DEEP LEARNING MODEL FOR IOE EDGE APPLICATION

Aim:

To implement and evaluate a Convolutional Neural Network (CNN)-based deep learning model for Internet of Everything (IoE) edge applications and analyze its performance using accuracy, loss, and a confusion matrix.

Objectives :

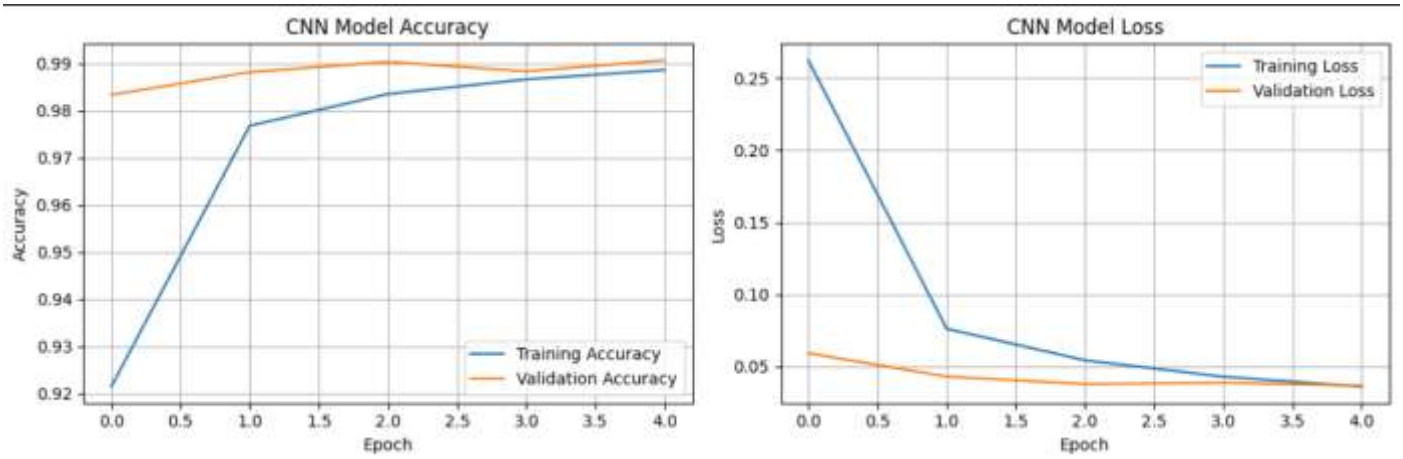
1. To load and preprocess the MNIST handwritten digit dataset for classification.
2. To develop and train a CNN model to recognize handwritten digits from 0 to 9.
3. To evaluate model performance using accuracy, loss, and a confusion matrix.

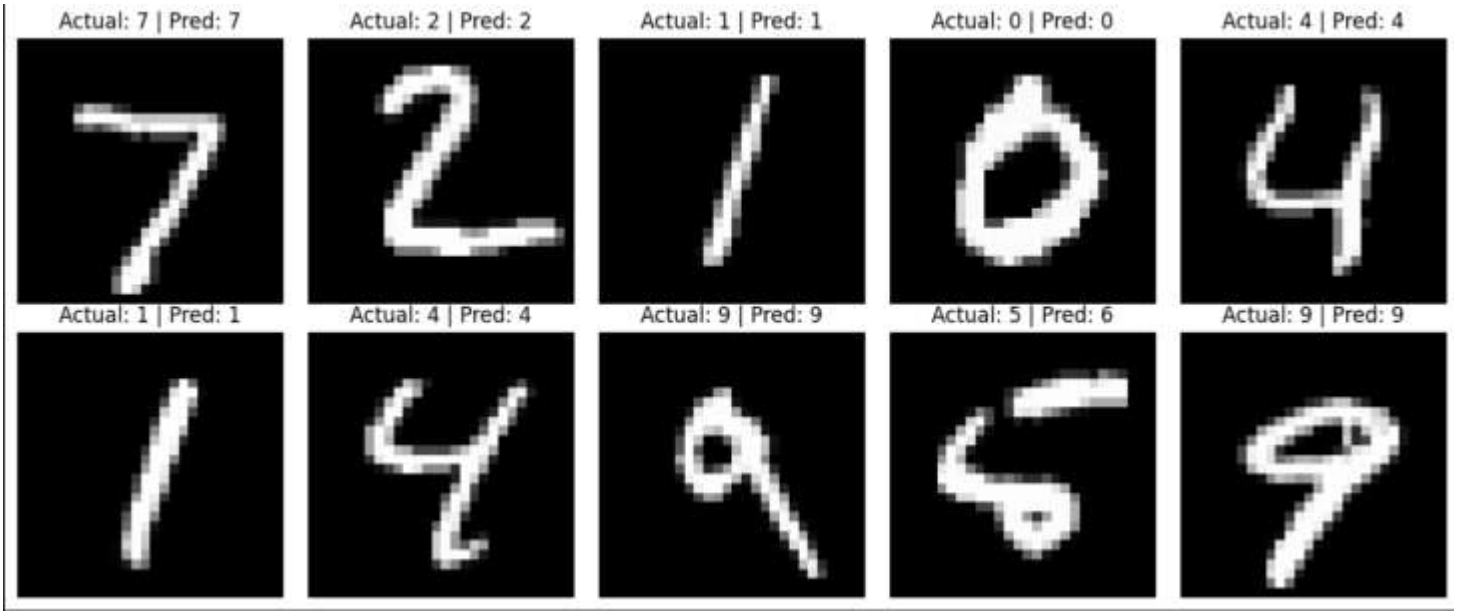
Procedure:

1. Open Google Colab and import the required Python libraries.
2. Load the MNIST dataset and preprocess the images.
3. Build and compile the CNN model.
4. Train the model using the training dataset.
5. Evaluate performance using accuracy, loss, and confusion matrix.
6. Display the results and predicted digits.

DIGIT IMAGE	ACCURACY	TESTING TIME
0	99.20	0.045 ms
1	99.50	0.042 ms
2	98.50	0.044 ms
3	98.70	0.043 ms
4	98.70	0.044 ms
5	98.20	0.043 ms
6	99.00	0.042 ms
7	98.70	0.044 ms
8	97.80	0.043 ms
9	97.90	0.044 ms

OUTPUT:





CONFUSION MATRIX:

